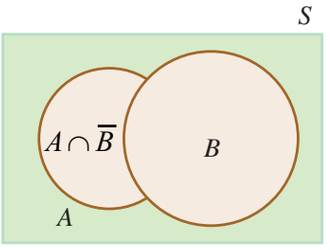
Theorem 12.6 (Addition theorem on probability)

If A and B are any two events, then

$$\boxed{P(A\cup B)=P(A)+P(B)-P(A\cap B)}$$



Proof

From the diagram,

$$A\cup B = (A\cap \overline{B})\cup B$$

$$P(A\cup B) = P\left[(A\cap \overline{B})\cup B\right]$$

$$= P(A\cap \overline{B})+P(B) \quad \text{(since } (A\cap \overline{B}) \text{ and } B \text{ are mutually exclusive)}$$

$$= [P(A)-P(A\cap B)]+P(B)$$

Therefore, $P(A\cup B) = P(A)+P(B)-P(A\cap B)$ ∎

Note 12.4

The above theorem can be extended to any 3 events.

(i) $P(A\cup B\cup C) = \{P(A)+P(B)+P(C)\}$

$$-\{P(A\cap B)+P(B\cap C)+P(C\cap A)\}+P(A\cap B\cap C)$$

(ii) $P(A\cup B\cup C) = 1-P(\overline{A\cup B\cup C}) = 1-P(\overline{A}\cap\overline{B}\cap\overline{C})$

Example 12.14

Given that $P(A)=0.52$, $P(B)=0.43$, and $P(A\cap B)=0.24$, find

(i) $P(A\cap \overline{B})$ (ii) $P(A\cup B)$ (iii) $P(\overline{A}\cap\overline{B})$ (iv) $P(\overline{A}\cup\overline{B})$.

Solution

(i) $P(A\cap \overline{B}) = P(A)-P(A\cap B)$

$$= 0.52-0.24 = 0.28$$

$$P(A\cap \overline{B}) = 0.28.$$

(ii) $P(A\cup B) = P(A)+P(B)-P(A\cap B)$

$$= 0.52+0.43-0.24$$

$$P(A\cup B) = 0.71.$$

(iii) $P(\overline{A}\cap\overline{B}) = P(\overline{A\cup B})$ (By de Morgan's law)

$$= 1-P(A\cup B)$$

$$= 1-0.71 = 0.29.$$

(iv) $P(\overline{A}\cup\overline{B}) = P(\overline{A\cap B})$ (By de Morgan's law)

$$= 1-P(A\cap B) = 1-0.24$$

$$= 0.76.$$

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